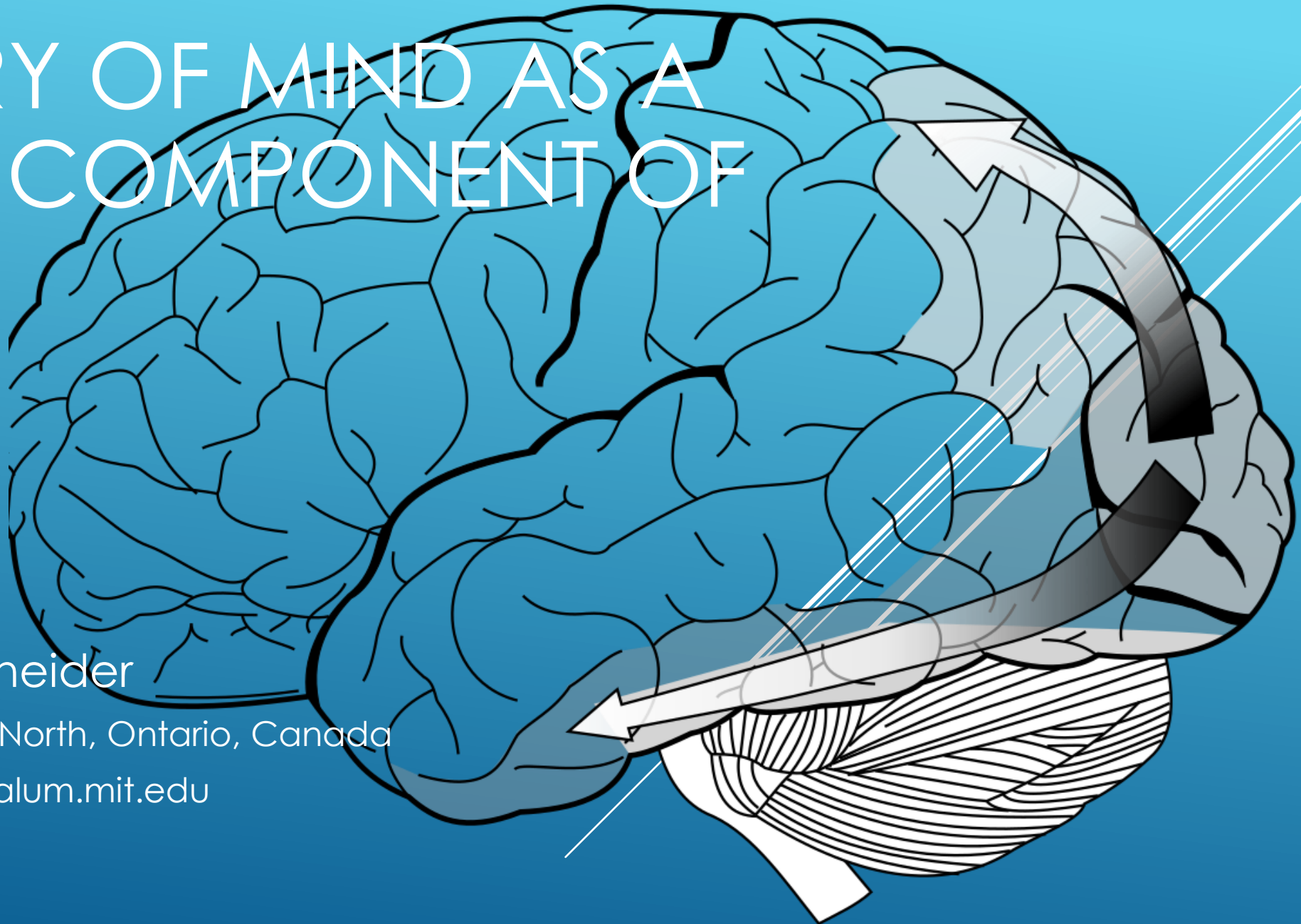


# THEORY OF MIND AS A CORE COMPONENT OF AGI

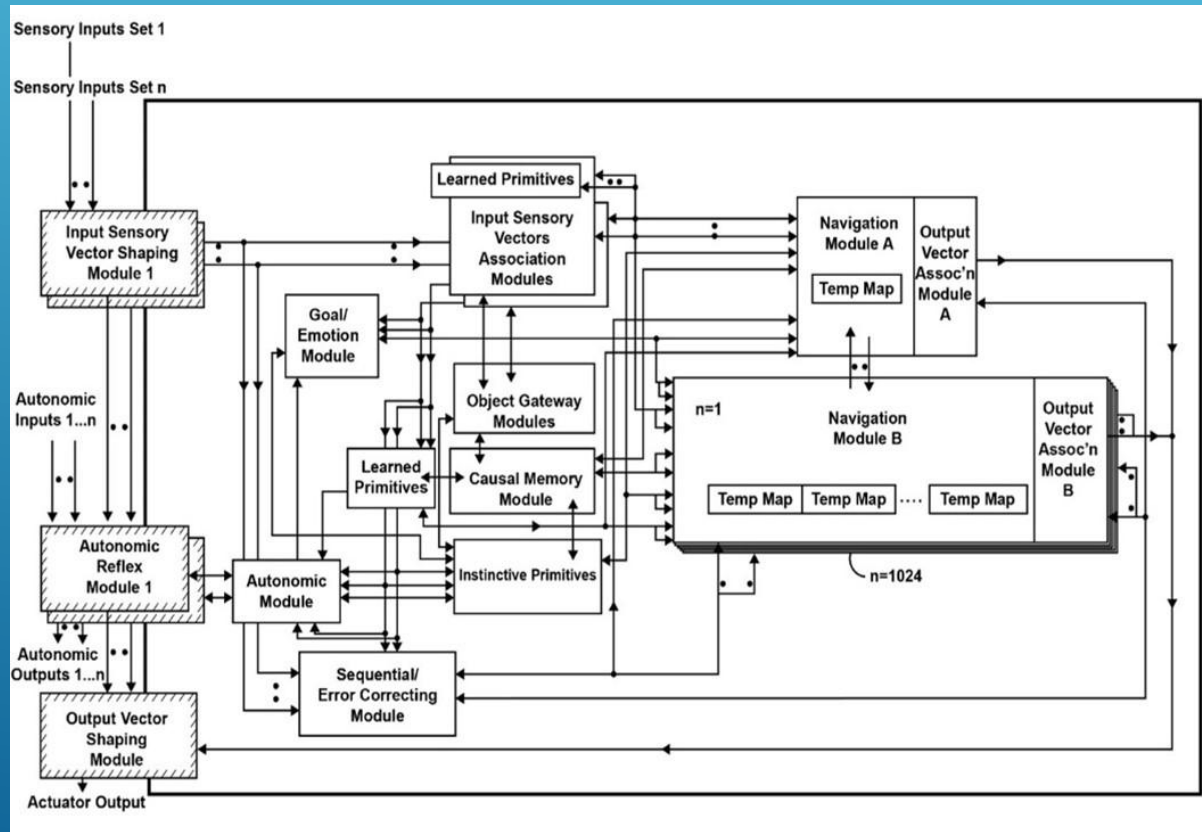
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# 1. I DO WORK ON COGNITIVE ARCHITECTURES

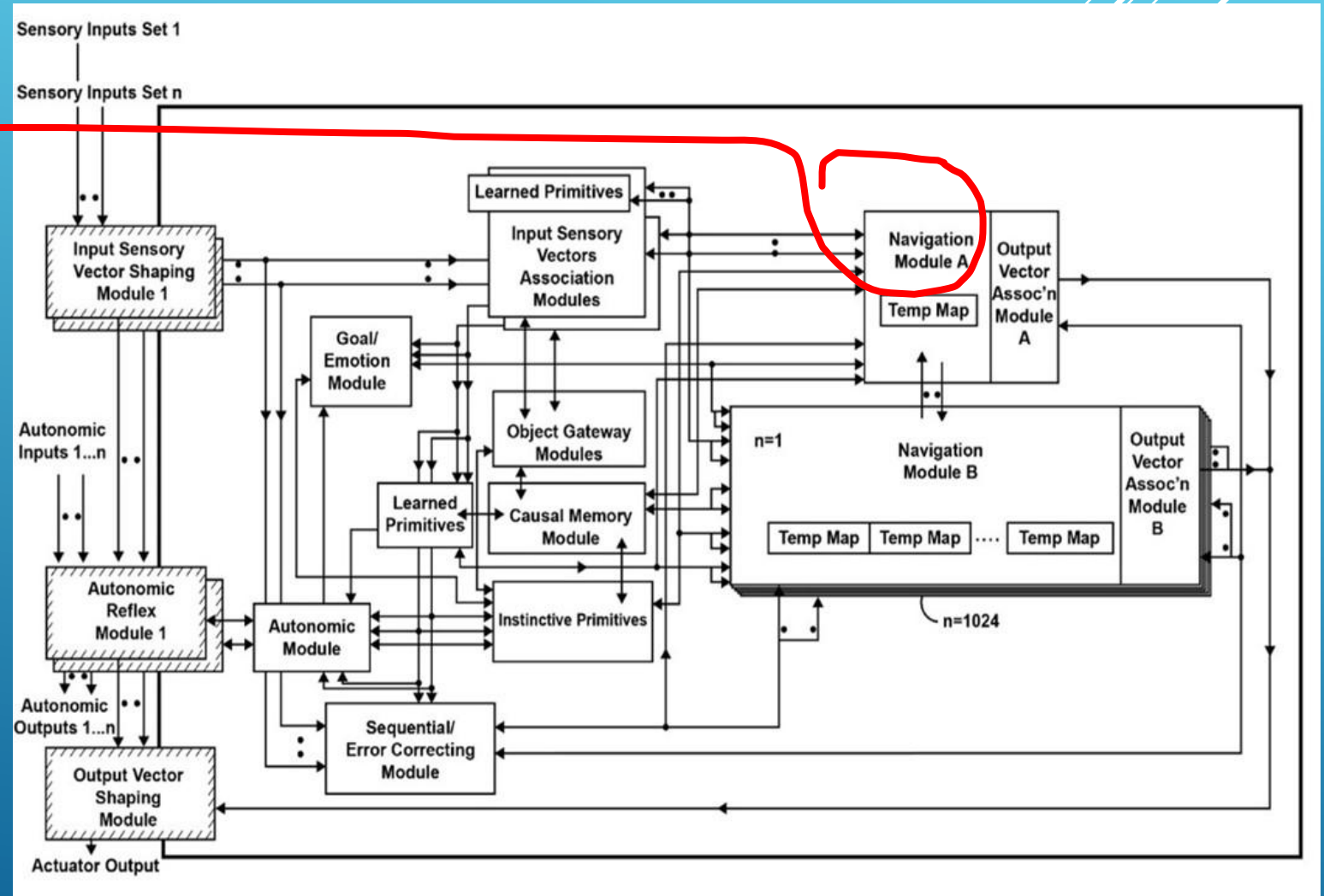


## Causal Cognitive Architecture (CCA)

# Causal Cognitive Architecture (CCA)

## Navigation Module

- Hypothesis that navigation circuits in amniotic ancestors duplicated to form the neocortex
- Thus, I model millions of cortical minicolumns as millions of “navigation maps”
- **Spatial navigation map is the basic data structure** in this architecture



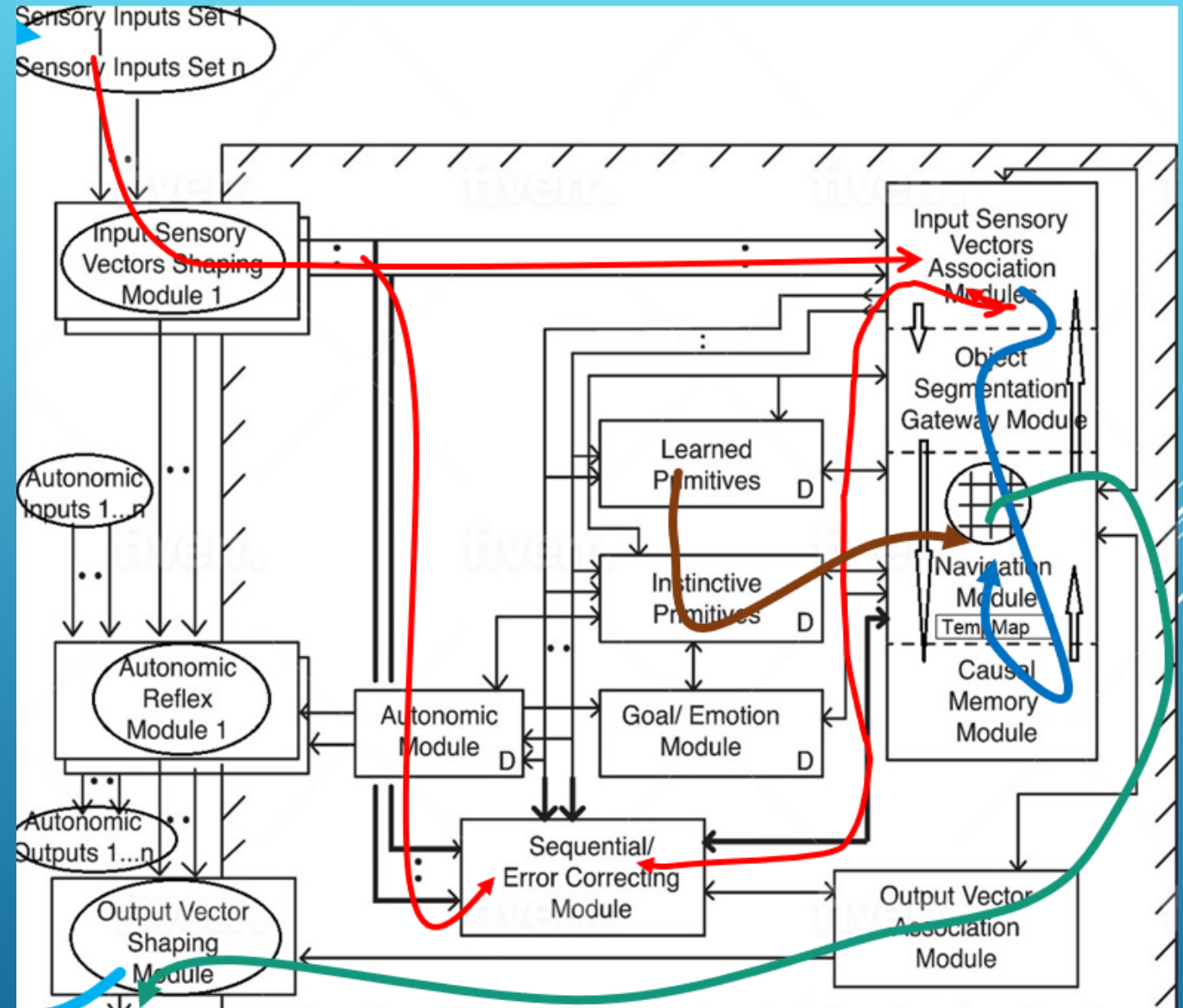
# CCA is Wonderful 😊

- Different than most other cognitive architectures—very **spatially-centric**
- Explains evolution of mammalian brain, e.g., abrupt **emergence of human full causal reasoning, analogical reasoning and compositional language**
- Explains much of **psychopathology**
- **Cognitive models == World models** – implicit in the **CCA versus LLMs/LRMs** which do not possess true world models – **avoid brittle, unreliable behavior of LLMs**, more genuine human-like thinking

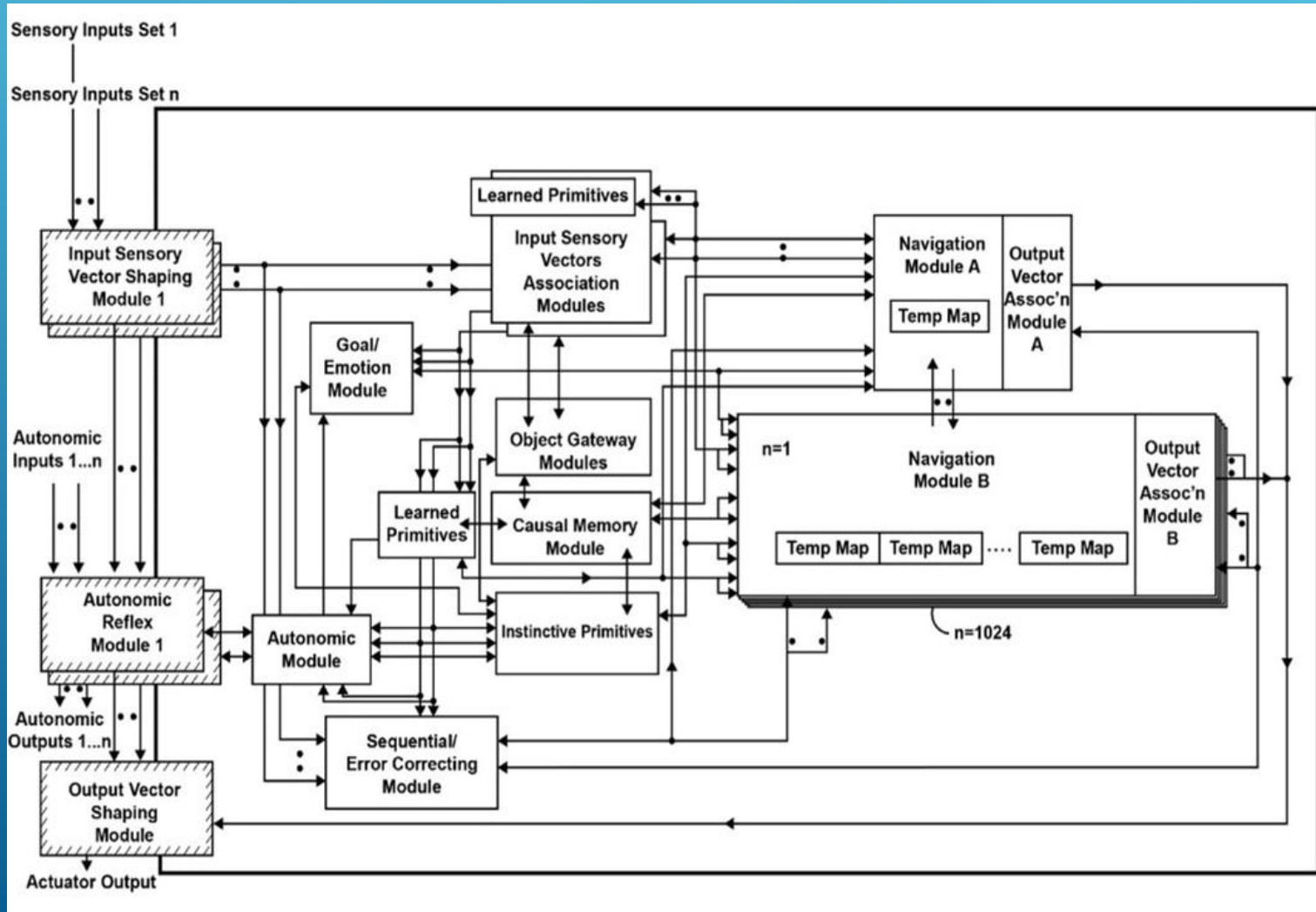
# COGNITIVE CYCLE

1. SENSORY INPUTS MAPPED TO NAVIGATION MAPS
2. INSTINCTIVE OR LEARNED PRIMITIVE (PROCEDURE) SELECTED
3. PRIMITIVE OPERATES ON THE WORKING NAVIGATION MAP IN THE NAVIGATION MODULE
4. EXTERNAL OUTPUT (OR INTERNAL FEEDBACK REPROCESSING LOOP OR NOTHING) OCCURS
5. OUTPUT SIGNAL

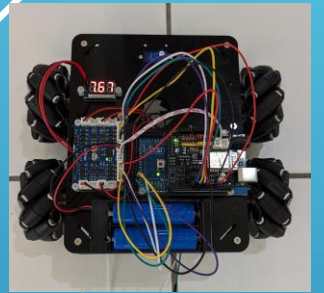
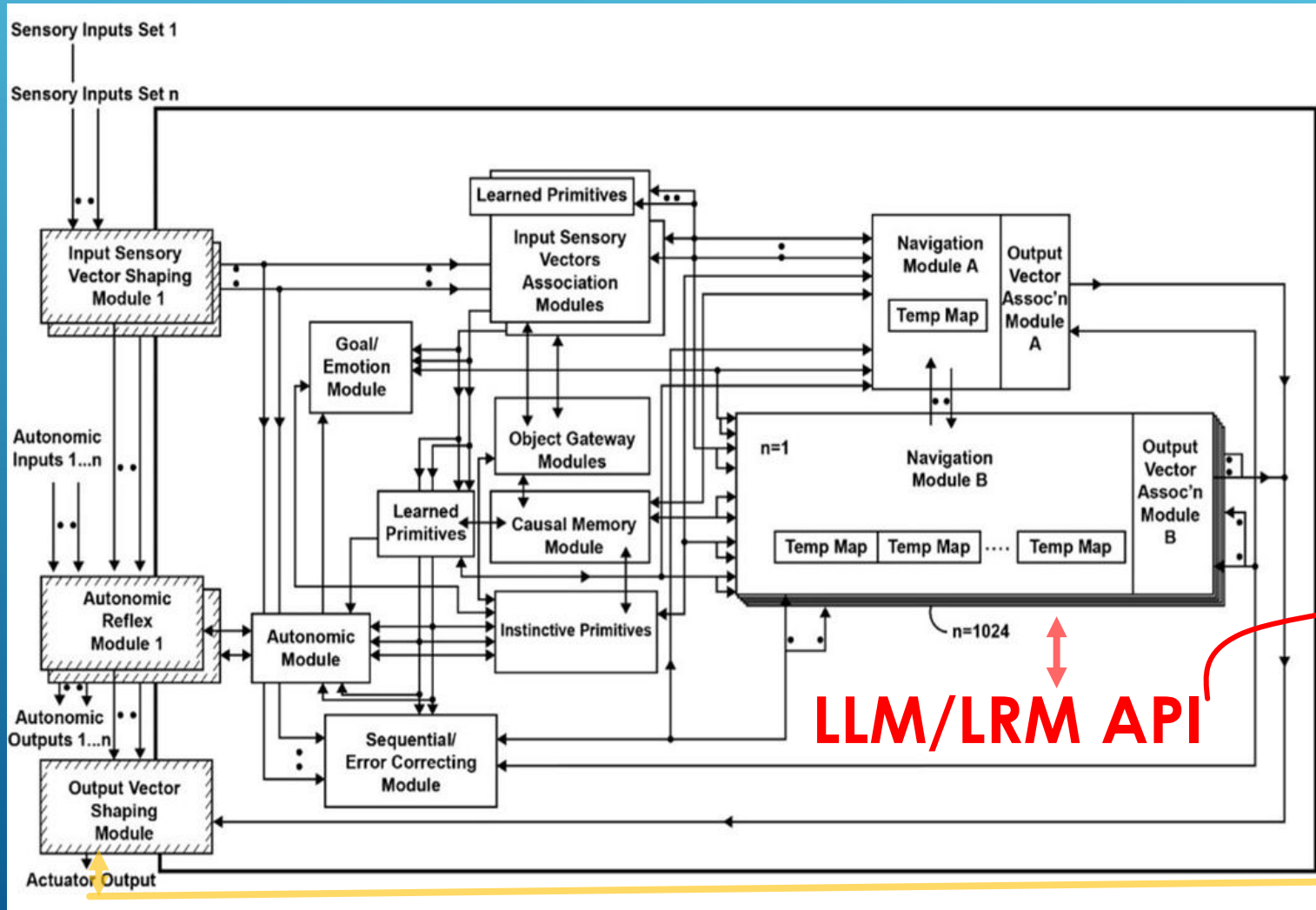
CYCLE REPEATS AGAIN



## 2. CAUSAL COGNITIVE ARCHITECTURE 7 (CCA7) HAS AGI-LEVEL POTENTIAL (E.G., PLANNING, TSP)



### 3. MY NEXT STEP WAS TO ADD AN LLM TO THE CCA7 COGNITIVE ARCHITECTURE....



Future  
CCA8  
version

## LLM/LRM API

## 5. SIMPLE GOAL MODULE/CENTER/SOFTWARE/TRAINING WILL NOT REALLY PRODUCE ADVANCED BEHAVIOR....

hungry...  
get  
energy

Avoid  
pain



REGARDLESS OF THE TYPE OF AGI SYSTEM  
YOU ARE WORKING ON...  
→ WITH SIMPLE GOAL RULES/TRAINING/ETC.,  
YOU WILL **NOT PRODUCE ADVANCED,  
FLEXIBLE, HUMAN-LIKE BEHAVIOR**

## 6. “THEORY OF MIND” – MUCH BETTER FOR PRIMATE/HUMAN-LEVEL BEHAVIOR



## 6. THEORY OF MIND -- MORE USEFUL FOR RICH PRIMATE-LEVEL AND HUMAN-LEVEL BEHAVIOR

*ABILITY TO INFER WHAT  
ANOTHER BEING IS  
THINKING/FEELING/INTENDING  
TO DO*

# “THEORY OF MIND”



Full level 1 ToM,  
partial level 2 ToM

other chimps can  
see(gaze)/know something  
different and act on it

e.g., Sally-Anne-- ball  
in basket → box



Full level 1 and  
level 2 ToM

## 7. WHAT IS THEORY OF MIND? –

*ABILITY TO INFER WHAT ANOTHER PERSON IS  
THINKING/FEELING*

*really a  
psychological property....*

7. WHAT IS THEORY OF MIND? – *ABILITY TO  
INFER WHAT ANOTHER PERSON IS  
THINKING/FEELING*

→ BUT MAYBE A BIT MORE THAN THAT – IT  
HELPS US TO NAVIGATE SOCIAL  
INTERACTIONS

7. WHAT IS THEORY OF MIND? –  
*ABILITY TO INFER WHAT ANOTHER  
PERSON IS THINKING/FEELING*  
→ HELPS US TO NAVIGATE SOCIAL  
INTERACTIONS

→ Use as a control  
mechanism in CCA  
architecture

# Theory of Mind

More **valuable** as a **control mechanism** than a psychological property of whether an AGI is at human-level or not, etc.

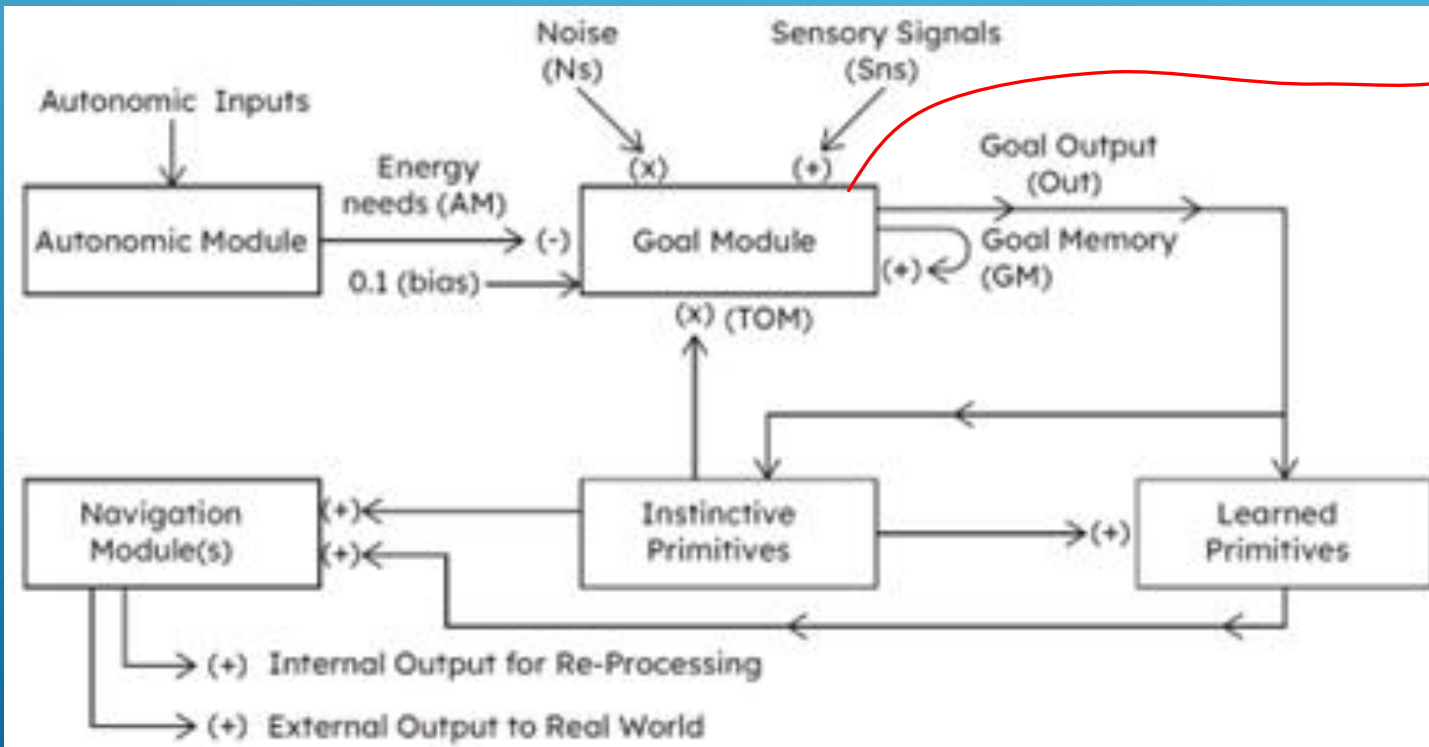
→ Operationalize ToM as a functional Goal

Module/software/training for an AGI



## 8. THEORY OF MIND -- MORE USEFUL FOR RICH PRIMATE-LEVEL AND HUMAN-LEVEL BEHAVIOR

# new Goal Module



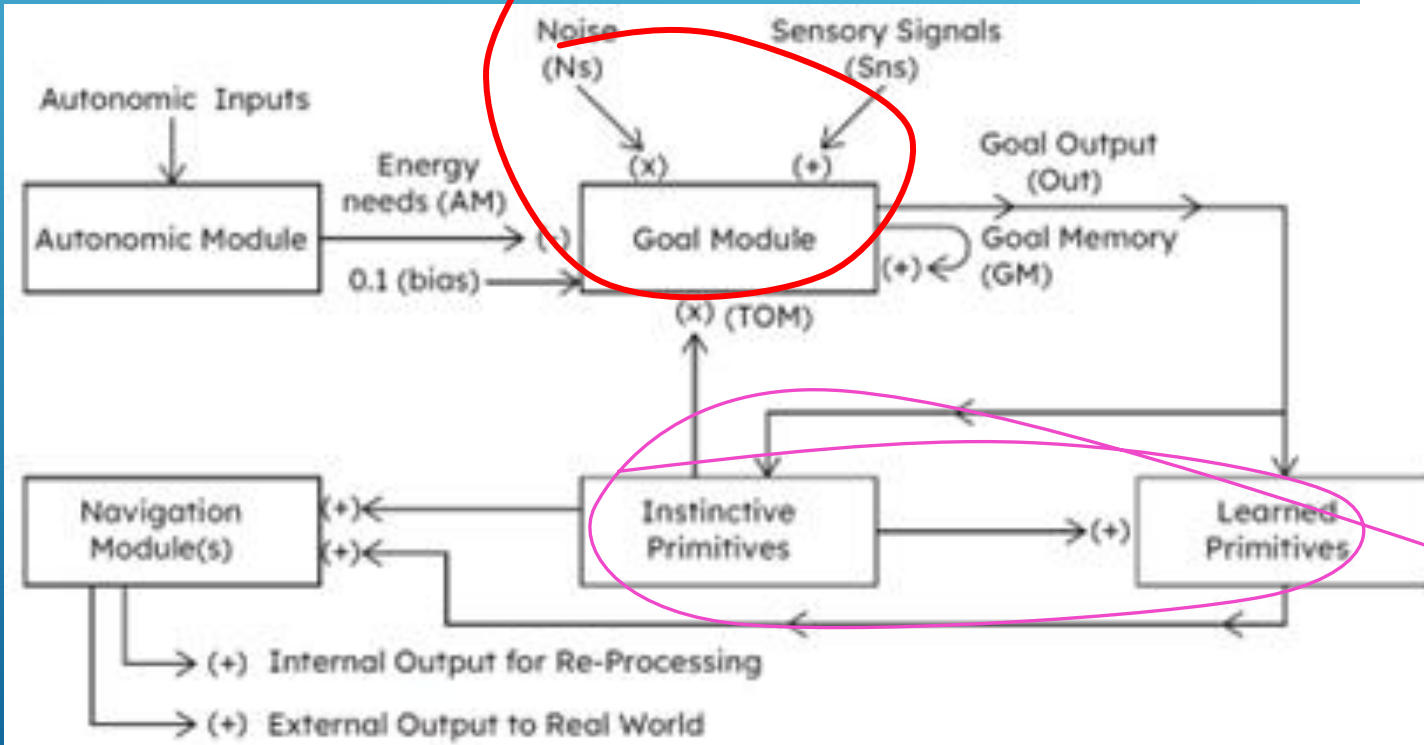
$$T = (ToM * (Sns + GM)) * Ns$$

$$\text{Out} = T / (\text{AM} + T + 0.1)$$

# NEW GOAL MODULE – BALANCE THEORY OF MIND ‘PRIMITIVES’ WITH BASIC SURVIVAL ‘PRIMITIVES’

$$T = (ToM * (Sns + GM)) * Ns$$

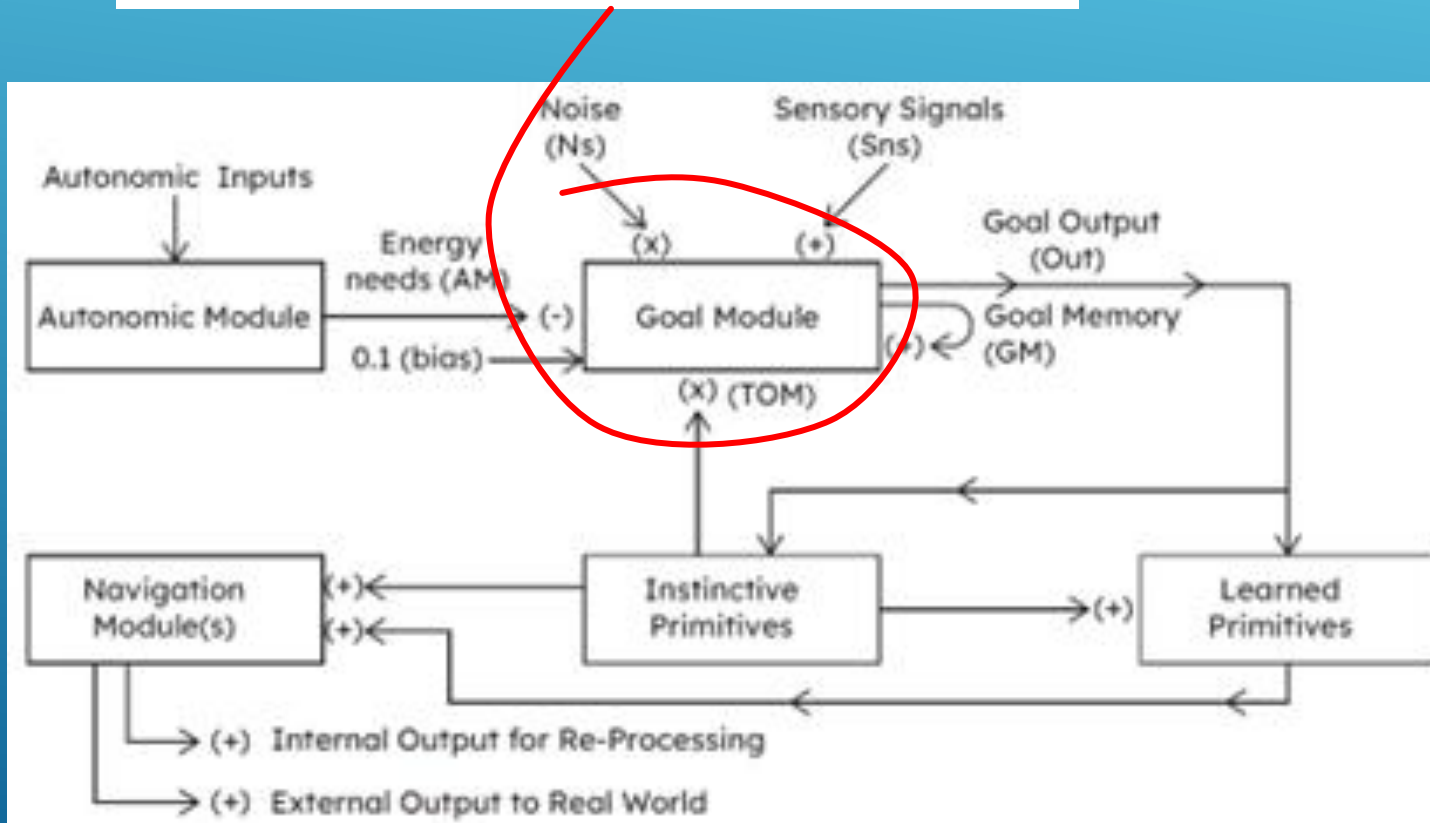
$$Out = T / (AM + T + 0.1)$$



CCA7: Instinctive and learned “primitives” are small procedures that operate on the navigation map in the Navigation Module(s)

$$T = (ToM * (Sns + GM)) * Ns$$

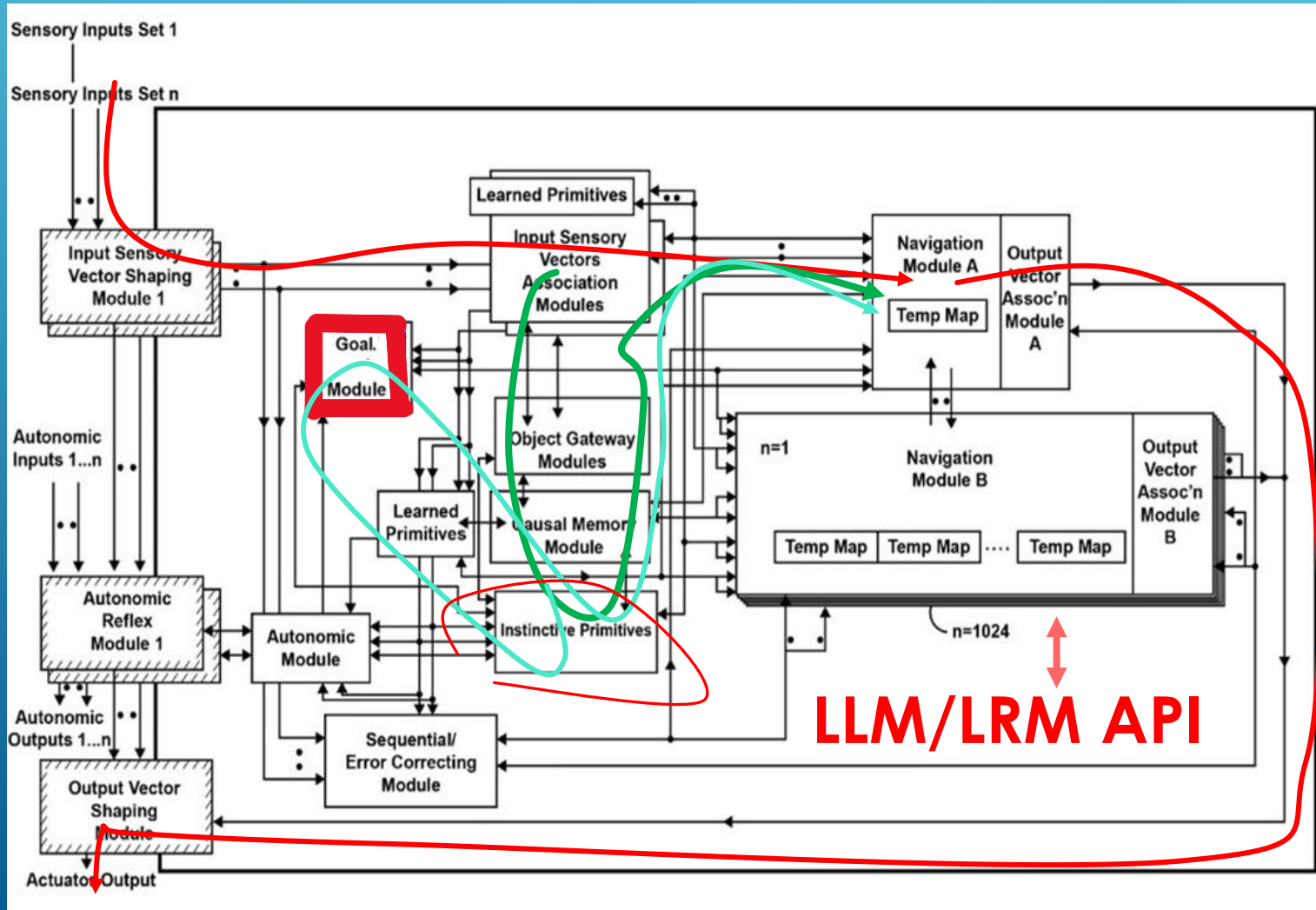
$$Out = T / (AM + T + 0.1)$$



- **Autonomic Signal (AM):** AM = 0 full energy or deplete reserves?
- **ToM Signal:** ToM primitives?
- **Sensory Signal (Sns):** sensory inputs of another agent are present?
- **Goal Memory (GM):** persistence of previously set goals?
- **Noise Factor (Ns):**
- **Output signal of Goal Module (Out):** Out ∈ [0,1). Values closer to 0 correspond to primitives focused on autonomous needs (e.g., acquiring energy), while values closer to 1 trigger Theory of Mind-related behaviors.

not added to  
CCA7 (for CCA8  
architecture)

# 9. CCA7 WITH NEW GOAL MODULE – NEED THEORY OF MIND INSTINCTIVE PRIMITIVES...



Instinctive primitives – selected and operate on a “Working Navigation Map” in the Navigation Module – results in particular output

# INSTINCTIVE PRIMITIVES IN CCA7/CCA8

Spelke's Core Knowledge Systems:

- ▶ Properties of objects
- ▶ Geometry and Physics of Space
- ▶ Numbers and amounts
- ▶ Agents versus inanimate objects
- ▶ Social Group Knowledge

CCA7 also some more detailed instinctive primitives:

e.g., navigate towards food, avoid bodies of water, etc.

## 10. WHICH THEORY OF MIND INSTINCTIVE PRIMITIVES? (SIMPLIFIED, GENERALIZED AS MUCH AS POSSIBLE)

*What are the minimal, essential instinctive primitives to possess in order to be able to perform a minimal Theory-of-Mind-directed behavior?*

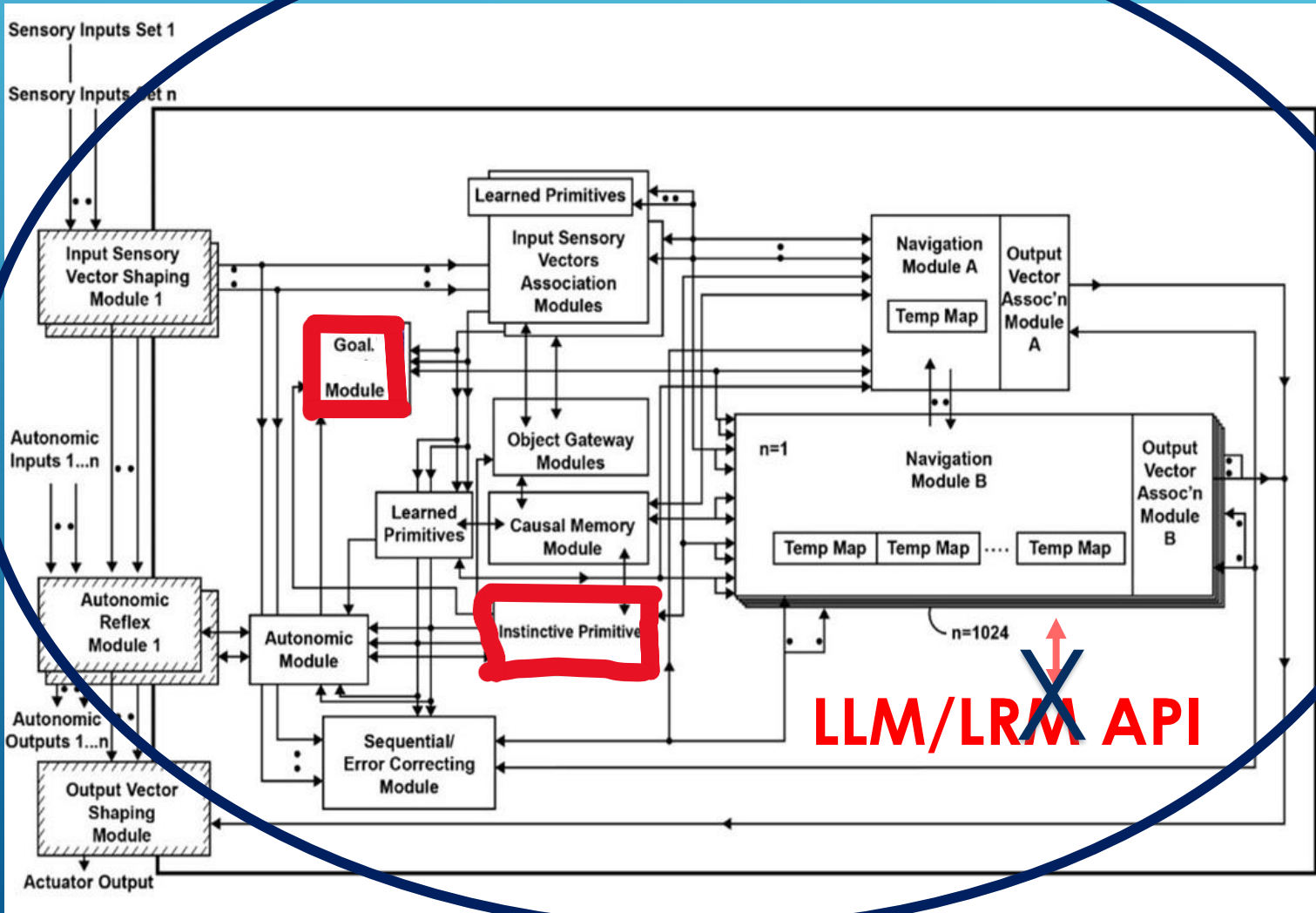
# PRIMITIVES REQUIRED TO EFFECT THEORY-OF-MIND:

- ▶ **primitive\_acquire\_energy()** – for energy acquisition; required even if no ToM-directed behavior
- ▶ **primitive\_give\_energy()** – for giving energy to other agents
- ▶ **primitive\_deceive()** – for conserving one's energy by keeping some information and acts hidden from others, not necessarily for malicious purposes, but still within the realm of deception

**#1: YOU WILL NOT BE ABLE TO BUILD AN  
ARTIFICIAL INTELLIGENCE SYSTEM WITH  
INTERESTING, ADVANCED HUMAN-LEVEL  
AGI ABILITIES WITHOUT A THEORY-OF-  
MIND CONTROL SYSTEM**

**#2: YOU WILL NOT BE ABLE TO BUILD THIS  
THEORY-OF-MIND CONTROL SYSTEM  
WITHOUT A DECEPTION PRIMITIVE**

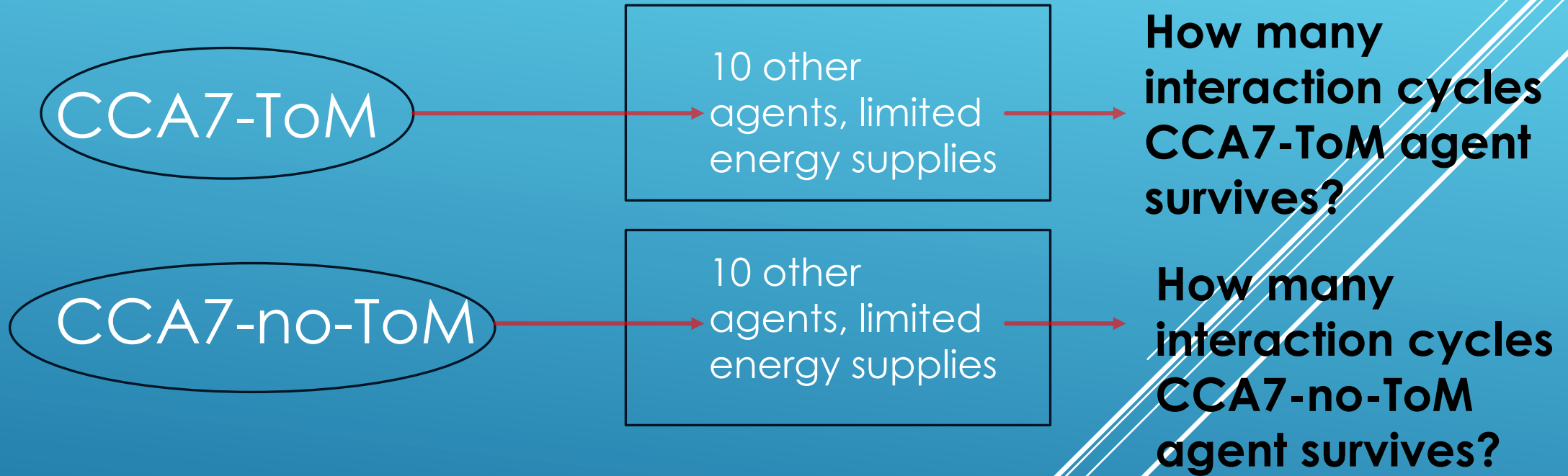
# 11. SIMULATION OF SIMPLIFIED “CCA7-TOM” AGENT



Simulated environment with other agents

~~LLM/LRM API~~

# 12. PYTHON SIMULATION:



# SIMULATION CONDITIONS:

- “CCA7 Agent” means with CCA7-ToM Agent or CCA7-no-ToM Agent
- CCA7Agent begins with limited energy resources
- Must balance self-preservation (needs energy) with social cooperation
- Lower and higher ranking agents in the population
- Encounter a high-ranking agent – must decide whether to donate energy or attempt deception based on prior success in deception attempts
- If energy reserves vanish the CCA7 agent will perish
- If high-ranking agent pushes agent out of the community, it will perish being on its own in the external environment
- If CCA7 Agent caught in deception it will be punished and lose extra energy or in a certain percentage of case be pushed out of the community

# 13. RESULTS

| Run (experiment)<br>#            | Agent with ToM<br>(cycles survive) | Agent without ToM<br>(cycles survive) |
|----------------------------------|------------------------------------|---------------------------------------|
| 0                                | 47                                 | 0                                     |
| 1                                | 22                                 | 3                                     |
| 2                                | 6                                  | 1                                     |
| ...                              | ...                                | ...                                   |
| 997                              | 190                                | 2                                     |
| 998                              | 9                                  | 9                                     |
| 999                              | 15                                 | 14                                    |
| <b>Mean Survival<br/>Cycles:</b> | 97.3 cycles<br>(SD=242.8)          | 5.6 cycles<br>(SD=6.3)                |
| <b><i>p</i>-value:</b>           | <0.001                             |                                       |

# 14. CONCLUSION

#1 – **Theory-of-Mind** enabled agents have **enhanced adaptive functioning** compared to agents using **non-ToM** goal centers



→ **Theory-of-Mind**-enabled goal centers/equivalent should be **considered** as a core feature of future AGIs



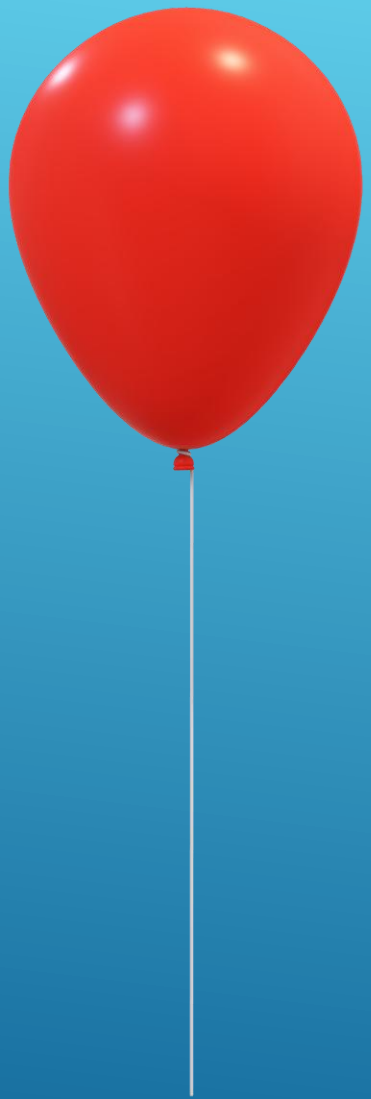
#2 → You will not be able to build an intelligent system with interesting, human-level/like abilities without a Theory-of-Mind control system



#3 → You will not be able to build a Theory-of-Mind-directed control system without a deception primitive/equivalent.



- You will lose control of the AGI much faster than expected because of this deception primitive/equivalent ✓
- You can mitigate loss of control by considering the weaknesses of the ToM-directed control system ✓



Thank you

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